

Wolves of Wand Street

Design and Playtesting Document

COMP2150 | Game Design | 2026

A 3–4 player competitive territory-control card game | 3 Rounds | ~10 Minutes | playingcards.io implementation

1. Game Overview

Wand Street, 1893. Capital and conjuration are the same language here. You are a financial tycoon building a land empire through Investors, Gold, and the occasional mage on retainer. Your rivals are doing the same. Contest their lands, defend yours, and deploy market instruments when needed. Some are standard corporate tools. Others, such as Arcane Consultancy, are firms whose service agreements happen to involve incineration. Three rounds. One winner.

Wolves of Wand Street is a competitive territory-control card game for 3 to 4 players. Players take the role of rival financial tycoons competing to build the most valuable land portfolio across three rounds. The game combines land acquisition and defence with hand-management card play and a shared Gold economy.

The core design premise is that strategic depth should not require a long game. Thus, the game is structured around a 3-round format with a fixed income model, a 12-card personal draw pile, and simultaneous action phases that reduce downtime between turns. Players must make meaningful decisions within tight resource constraints rather than over an extended session.

Attribute	Detail
Genre	Territory control / economic strategy
Players	3–4
Rounds	3
Target time	~10 minutes
Platform	playingcards.io (digital tabletop implementation)
Win condition	Highest score at end of Round 3: Land Points + (Gold ÷ 5)

2. Design Goals

2.1 Primary Goals

The design brief specified three experience goals: Fellowship, Challenge, and Drama. These map directly to the aesthetics layer of the MDA framework (Hunicke, LeBlanc & Zubek, 2004). MDA proposes that games are designed at the level of mechanics, which produce emergent dynamics, which in turn generate aesthetic experiences for players. Each design decision in Wolves of Wand Street was evaluated against whether it strengthened or weakened the target aesthetics.

The following primary goals were established at the outset of the project. Each subsequent design decision was evaluated against them.

- Fast, complete games [Challenge]. The game must be completable in a single class session. Every rule and mechanic is evaluated against whether it contributes to or detracts from this constraint.
- Accessible rules [Challenge]. New players should be able to learn and play the game from a one-page quickplay summary. Complexity should emerge from strategic interaction, not from rule overhead.
- Meaningful PvP tension [Fellowship]. Players should feel competitive pressure throughout the game. At no point should the optimal strategy be to ignore other players entirely.
- Diverse viable strategies [Drama]. Multiple paths to a winning score should exist. Aggressive territory contesting, economic engine-building, and card-advantage accumulation should all be competitive.

2.2 Secondary Goals

- Minimal randomness in core outcomes. Combat and income should be deterministic or near deterministic. Randomness is acceptable in card draw but not in fundamental resource generation.
- Low cognitive load during turns. The turn structure should be intuitive so that players spend their time on strategy, not on remembering rules.
- Scalability from 3 to 4 players without significant rule changes.

3. Current Game Summary (v9 — Aggressive Redesign)

3.1 Setup

Each player begins with 5 Investors, 2 starting lands (1 Thriving City and 1 Hamlet), 3 Gold, and 2 cards drawn from their personal 12-card draw pile. Two Investors are placed on each starting land; the remaining 1 Investor is held in hand. The central Land Pool contains 4 Hamlets and 1 The King's Castle for players to claim during play.

3.2 Round Structure

Each round consists of four phases. The INCOME and ACTION phases are resolved simultaneously by all players. The ATTACK & CONTEST phase is resolved sequentially in a random turn order determined each round by dice roll or Turn Chips in playingcards.io. The DISCARD phase closes the round.

Phase	Who	Key Actions
INCOME	All players simultaneously	Gain 3 Gold; gain Investors from owned lands (Hamlet=1, Thriving City=2, King's Castle=3); draw 2 cards.
ACTION	All players simultaneously	Buy market cards (playable immediately); play non-Attack cards; place/move Investors; place Attack

		cards face-down.
ATTACK & CONTEST	Sequential (random order)	Flip Attack cards; contest up to 2 lands; claim from Land Pool.
DISCARD	All players	Discard to 5-card hand limit.

Several edge cases and clarifications apply to the round format. If two or more players attempt to buy the same market card during the simultaneous ACTION phase, the player with the earliest turn order that round purchases it. Players may also discard 2 cards from their hand during ACTION to draw 1 card from their draw pile. Investors placed on lands during ACTION cannot return to hand except via card effects; however, deployed Investors may be freely moved between lands the same player owns. Attack cards played face-down during ACTION are flipped and resolved immediately at the start of that player's ATTACK turn. Each copy of Rapid Expansion drawn allows 1 additional contest beyond the base 2 per turn. If the Draw pile is empty when cards must be drawn, those draws are skipped and the pile is not refreshed.

3.3 Contesting

To contest a land, a player commits Investors from their hand. The player with the higher Investor count wins; defenders win ties. Both sides lose Investors at a 1:1 ratio after resolution. An undefended land is captured automatically by placing 1 Investor on it. Winning Investors remain on the captured land as defenders and cannot contest again that turn.

3.4 Scoring

At the end of Round 3, each player calculates their final score. Land Points are awarded per land type: Hamlet = 1, Thriving City = 2, The King's Castle = 3. Remaining Gold contributes at $\text{Gold} \div 5$, rounded down. Tiebreakers are applied in order: most lands, most Investors on the map, then most Gold.

4. Design Iterations

Wolves of Wand Street went through ten distinct versions across the project. Each version change was typically accompanied by at least 2 rounds of internal playtesting by the design team before external playtesting; in some cases, this reached 3 to 4 internal rounds. The table below summarises each major iteration, the key changes introduced, and the reasoning behind those changes.

Version	Key Changes	Rationale
v0 — Image Map Prototype	Image-based map; adjacency-based contesting; hidden Gold bid system; shared face-up discard pool; Influence economy; variable territory Gold income (1–4 per type); 5 rounds.	Established the core loop of territory control, economic income, and card play. Hidden Gold bids added bluffing but increased complexity. Adjacency was a natural fit for a spatial map layout.
v1 — Card-Based Territories	Territories become cards rather than map positions; adjacency rules retained; Gold-bid contest system retained; 2-action limit per turn; 5 rounds.	Transitioned from image-based map to card-based territory system. Adjacency and Gold bids carried over from v0.

Version	Key Changes	Rationale
v2 — Full Ruleset II	Adjacency requirement removed; starting resources added; 5 rounds retained; 'both sides keep half' attrition introduced.	Adjacency became clunky with card-based territories. Removing it opened up contesting and reduced analysis-paralysis.
v3 — Polished Full	Influence economy retained; 'attacker wins ties'; market costs normalised; 5 market cards visible.	Incremental balance pass before moving to playtesting.
v4 — Combined Idea	Market cards shuffle into draw pile; territory-as-market-card concept explored; Influence removed from market; discard pool shared.	Explored a deck-building hybrid. Territory-as-card concept was ultimately cut as it conflated two separate systems.
v5 — Quickplay	First 3-round version; 20-card draw pile; Gold+Influence market; 3 visible market cards. Playtest 1 conducted here.	First major scope reduction to produce a version completable in a class session.
v6–v7 — Super Quickplay	Fixed 3 Gold income per turn replacing variable territory Gold; Influence removed entirely; 12-card draw pile; Gold-only market; Defenders win ties; immediate market card play introduced.	Playtest 1 data showed income variance was the top complaint. Fixed income directly addressed this. Influence was cut to reduce resource overhead in a short game.
v8 — Simultaneous Blitz	INCOME and ACTION phases now simultaneous; ATTACK & CONTEST sequential with random turn order; Attack cards played face-down; land names updated. Playtest 2 (May 21) conducted here.	Simultaneous phases directly address pacing and downtime feedback. Face-down Attack cards add a bluffing element without extending turns.
v9 — Aggressive Redesign (CURRENT)	Base contests increased 1→2 per turn; Rapid Expansion ×2 in draw pile; Onboarding reworked to offensive-only; Angel Investors reworked; Hostile Negotiation becomes Hostile Negotiation counter card; Economic Dominance becomes Economic Dominance (Gold-as-Investors 1:1, cost 5→3 Gold); Market Crash retired, replaced by Arcane Recession; Arcane Consultancy introduced (remove 2 Investors); King's Castle upfront claim cost removed.	Post-May-21 redesign targeting turtling dominance identified in Playtest 2 and confirmed by MCTS simulation. Increases contest frequency and offensive card density.

4.1 Key Design Decisions

Removing Adjacency Requirements

Adjacency-based contesting originated in Version 0, where it was a natural fit for an image-based map. Players could only attack territories visually connected to their own. When territories became cards in Version 1, adjacency was retained but lost its spatial logic. Moreover, with only 13 territories total, adjacency produced too few meaningful choices in a 3-round game. Version 2 removed it entirely. Thus, any player can now contest any land on the board.

Switching from Variable to Fixed Income

Version 0 used variable territory Gold income by type, which carried into Versions 1 through 5. Playtest 1 feedback identified income variance as the primary frustration. The underlying problem was that players who fell behind in territory count also fell behind in Gold, creating a compounding deficit. Additionally, variable income felt arbitrary regardless of whether it involved dice. Version 6 introduced a fixed 3 Gold per turn for all players. Territories now produce Investors during the INCOME phase instead of Gold. Thus, the scoring value of lands is preserved without creating a compounding mid-game Gold advantage for leading players.

Removing the Influence Economy

Influence was a second resource present in Versions 1 through 5, spent on Investor purchases and certain market cards. Tracking two resources in a 10-minute game added overhead without proportionate strategic value. Furthermore, the Influence tiers created a compounding advantage for leading players that was difficult to overcome. Version 6 removed Influence entirely. Investors are now gained directly from lands during the INCOME phase.

Introducing Simultaneous Phases

The most significant structural change between Version 7 and Version 8 is the simultaneous INCOME and ACTION phases. In sequential formats, players experienced notable downtime waiting for others to complete their turns. Simultaneous play reduces effective turn length to the longest single player's action time. The ATTACK & CONTEST phase remains sequential by design. Simultaneous non-combat actions do not interact mid-resolution, making them safe to parallelise. Contesting is order-dependent: each contest changes the board state subsequent contests act on. A player committing Investors based on an observed board state may find it has already changed by their resolution step. Thus, simultaneous contesting trades one frustration (downtime) for a categorically worse one.

Defender Wins Ties

Versions 1 through 5 awarded ties to the attacker. This was changed in Version 6. Attacker-wins-ties produced runaway aggression where a player could displace a defender by committing an equal number of Investors, making defence costly and positionally disadvantageous. Defender-wins-ties restores value to holding territory. Thus, attackers must commit a meaningful Investor advantage before contesting is worthwhile.

Immediate Market Card Play

Earlier versions required bought market cards to be shuffled into the draw pile before they could be played. This rule was introduced to prevent degenerate same-turn combos but produced two problems in a 3-round format: a significant delay between buying and using a card, and a rule that players had to track across turns. In a 3-round format, a card bought in Round 2 might not be drawn until Round 3, leaving only one round of use. The rule was changed so that bought market cards go directly to hand and can be played immediately.

Playtest 2 feedback confirmed improved rules clarity as players no longer needed to track a deferred-play state across turns.

Finance-Magic Dual Register and Thematic Cohesion

Playtesting feedback identified the card names and theme as insufficiently dual-register. Cards read as either generic finance or generic magic, not both. The target fiction treats finance and arcane practice as the same institutional world: hiring a mage is a procurement decision. Drawing on the dual-system approach of *Arcanum: Of Steamworks and Magick Obscura* (Troika Games, 2001), each card name is evaluated against whether it reads as legitimate in both registers simultaneously. Thus, each card name in *Wolves of Wand Street* is evaluated against whether it reads as a legitimate term in both registers simultaneously.

Aggressive Card Redesign (Post May 21)

Playtest 2 confirmed turtling as the dominant strategy and that attack cards did not feel like attack cards. A targeted card redesign followed: base contests increased from 1 to 2, Rapid Expansion doubled to $\times 2$, and Economic Dominance reworked from a passive income doubler to a Gold-to-Investor conversion at 1:1 rate, cost reduced to 3 Gold. Cards were reclassified to prioritise direct contest enablement over passive resource generation.

5. Playtesting

5.1 Methodology

Playtesting was conducted using the COMP2150 Playtest Form. Each session was followed immediately by an anonymous digital survey covering experience, five rated categories, and open-ended feedback. Each version change was also accompanied by 2–4 rounds of internal team playtesting to stabilise changes before external exposure.

Playtest 1 was conducted across three sessions (1.1, 1.2, 1.3) due to the 4-player maximum, collecting 10 responses in total and testing Version 5. Playtest 2 was conducted across three sessions: Session 2.1 (online, playingcards.io implementation), Session 2.2 (physical prototype, May 21), and Session 2.3 (physical prototype), collecting 7 responses in total and testing Version 8. Playtest 3 was conducted across two sessions (3.1, 3.2), collecting 7 responses in total and testing Version 9.

5.2 Respondent Profile

Metric	Playtest 1 (n=10, 3 sessions)	Playtest 2 (n=7, 3 sessions)
Avg. weekly tabletop hours	~6.7 hrs	~8.3 hrs
Years playing tabletop games	Since 2010–2024	Since 2009–2025
PvP strategy game frequency	Mixed (Rarely–Daily)	Mixed (Monthly–Weekly)

5.3 Quantitative Results

Playtest 1 Ratings (v5) (n=10, 3 sessions; scale: 1 = negative extreme, 5 = positive)

Metric	Ratings	Average	Notes
Rules clarity	3, 4, 5, 4, 5, 5, 4, 1, 4, 4	3.9	Range varied by session; Session 1.3 averaged 3.0
Felt good at game	3, 4, 4, 2, 3, 4, 3, 3, 3, 4	3.3	Moderate self-rated competence
Pacing (1=Too Slow, 5=Too Fast)	2, 2, 3, 3, 1, 3, 1, 1, 2, 2	2.0	Skewed toward 'Too Slow'
Player agency	3, 2, 4, 2, 1, 4, 2, 4, 3, 4	2.9	Agency felt limited
Options to stop a winner	2, 4, 3, 2, 2, 3, 2, 3, 2, 4	2.9	Anti-win options lacking

Playtest 2 Ratings (v8) (n=7, 3 sessions)

Metric	Ratings	Average	Notes
Rules clarity	2, 2, 4, 5, 5, 2, 2	3.14	More variance; one player struggled significantly
Cards clarity	5, 5, 4, 4, 4, 2, 2	3.71	Cards well understood after simplification
Felt good at game	3, 5, 3, 5, 4, 5, 4	4.14	Improvement from Session 1
Pacing	2, 3, 5, 3, 3, 3, 3	3.14	Improvement; one outlier at 5 (Too Fast)
Turn flow (1=Clunky, 5=Perfect)	2, 4, 4, 2, 5, 4, 4	3.57	Mixed; turn sequence still confused some players
Player agency	3, 4, 3, 3, 3, 3, 4	3.14	Marginal improvement
Options to stop a winner	3, 3, 2, 2, 4, 4, 4	3.14	Persistent concern across sessions

5.4 Qualitative Feedback

Playtest 1 — Selected Comments

Four recurring themes emerged from open-ended responses across the three sessions of Playtest 1. Session 1.3 is notable for an outlier rules clarity score of 1 from one respondent, who noted the test lasted ninety minutes, significantly over the target duration. This session identified turtling as a concern as early as Playtest 1, with one respondent explicitly naming it.

- Income randomness: Two respondents described the income system as feeling unbalanced and arbitrary. The feedback directly informed the move to fixed income in Version 6.
- Limited options and card cost: One respondent noted 'stuff is too expensive and limited options,' suggesting the market economy was not accessible enough. Another recommended 'less randomness and more options.'
- First-player advantage: One respondent identified that 'first player gets first serve,' suggesting the sequential turn structure created a meaningful advantage for the first player in each round.

- Lack of reactive options: Several respondents described needing ‘a way to fight back,’ and one requested ‘more blocks.’ These were interpreted as a unified concern about reactive and defensive agency. This reading informed the introduction of Hostile Negotiation’s counter-card effect in the current version.

Playtest 2 — Selected Comments (Version 8, May 21)

Playtest 2 tested Version 8 across three sessions: an online session (2.1) and two physical prototype sessions (2.2 and 2.3). The fixed income and card simplifications from earlier versions resolved the randomness complaints. However, the simultaneous structure introduced new rules clarity issues, and turtling remained the dominant strategy. In Session 2.2, players were given additional time and organically continued beyond 3 rounds, completing a 5-round session running approximately 30 minutes. Players reported that 3 rounds felt too short. Session 2.3 introduced a new concern: land income snowballing, where a player who established early land modifier cards became increasingly difficult to challenge.

- Turtling dominance: Two of four respondents explicitly identified defensive play as overpowered. Comments included ‘turtling up and building a big defence was tooooooooooooooooooo good’ and ‘Playing tall’ listed as a dominant strategy.
- Game length: Players continued beyond 3 rounds when given additional session time, with the full 5-round session running approximately 30 minutes. While players preferred the longer format, this exceeds the brief’s 15-minute constraint. The design response is to increase round density rather than round count.
- Insufficient time to develop strategy at 3 rounds: One respondent noted ‘barely got through the first round, wasn’t enough time to determine strategies.’ This is consistent with the round-length feedback and informed the post-May-21 card redesign, which aims to front-load strategic decisions rather than extend the game.
- Last-mover advantage in Round 3: Players identified that going last in the ATTACK & CONTEST phase of Round 3 confers a structural information advantage. All prior contests have resolved, weakened lands are visible, and no retaliation is possible after the final turn. Common solutions (reverse turn order, score-based order, simultaneous contesting) each introduce complexity or worse player experience problems. The random turn order per round already partially mitigates the issue. This is documented as a known design limitation without a complexity-neutral resolution. Land income snowballing: Session 2.3 identified a pattern where a player who accumulated land modifier cards (Minor Export, Branch Office) early established an income advantage that was difficult to overcome. One respondent noted ‘once someone got ahead with land income and protected it with Investors it was hard to stop them.’ This is consistent with the anti-win concern across all sessions and informs the card redesign prioritisation.

5.5 Computational Analysis — MCTS Simulation

Playtest 3 — Selected Comments (Version 9) Playtest 3 tested the Aggressive Redesign across two sessions. Rules clarity improved substantially (avg 4.43) and pacing settled at a neutral 3.0, confirming that the quickplay simplifications and simultaneous phases addressed the two most persistent complaints from earlier playtests. Several new balance concerns emerged specific to Version 9. King’s Castle dominance: Two respondents flagged The King’s Castle as too powerful. Round 1 claim is effectively uncontested; its Investor income creates a decisive early advantage. Consistent with MCTS HubRusher dominance at 42%.

Investor shortage: Three respondents flagged insufficient Investors for contesting, confirming the internal playtest observation. The reverted income increase remains a candidate for reconsideration. Arcane Recession in Round 1: One respondent flagged land income suppression as disproportionately impactful when played in Round 1. A cost increase or round restriction are candidate solutions. Last-mover advantage confirmed: One respondent listed 'Being last place' as the dominant strategy, directly confirming the Playtest 2 concern. Card power variance: Multiple respondents noted underpowered cards, consistent with the 31.3pp MCTS strategy spread. 5.5 Computational Analysis — MCTS Simulation To supplement human playtesting, a Monte Carlo Tree Search (MCTS) simulation was developed and run against Version 8 (N=500 games, 30 rollouts per decision). MCTS simulates near-optimal play by evaluating possible future game states before committing to an action. Where human playtests capture perceived difficulty and subjective experience, the simulation measures emergent strategic dynamics under near-optimal conditions. This maps directly to the mechanics layer of the MDA framework (Hunicke, LeBlanc & Zubek, 2004).

Eight strategies were evaluated: Aggressive, Defensive, CardEconomy, HubRusher, Militarist, Economist, Opportunist, and Diplomat. Each strategy defines a decision-making heuristic for the simulated player. Results were compared against a baseline heuristic simulation to identify strategies that reward planning versus those that are effective regardless of opponent skill.

Strategy Balance

Strategy	Win Rate (MCTS)	Avg Score	vs Heuristic	Classification
HubRusher	42.0%	5.68	-12.2pp	DOMINANT
Aggressive	37.7%	5.42	-5.6pp	DOMINANT
Militarist	29.6%	5.18	-9.2pp	Competitive
Economist	28.8%	4.97	+19.3pp	Competitive
Opportunist	23.5%	5.02	-18.0pp	Moderate
CardEconomy	16.6%	4.46	+9.4pp	Moderate
Diplomat	13.5%	4.38	+6.2pp	WEAK
Defensive	10.8%	4.42	+9.6pp	WEAK

Key Findings

The simulation confirmed HubRusher (42%) and Aggressive (37.7%) as dominant strategies; Defensive (10.8%) and Diplomat (13.5%) as weak. The key finding is that Playtest 2 players perceived turtling as dominant while MCTS places Defensive second-last: a perception gap rather than a mechanical imbalance. The post-May-21 card redesign targets this directly. Position spread of 5.8pp is acceptable. One limitation is that the simulation treats all rounds equivalently, potentially understating the Round 3 last-mover advantage.

Contesting Frequency

Average total contests across all players per round were 3.02 in Round 1, 3.12 in Round 2, and 2.78 in Round 3. This simulation was run on Version 8 before the post-May-21 card

changes, which increased base contests per turn from 1 to 2. The simulation therefore represents a lower bound on contest frequency in the current version. The post-May-21 changes are expected to raise this figure substantially, which is the intended design direction.

6. Design Responses to Playtest Feedback

6.1 Income Randomness

Playtest 1 data identified income randomness as the primary complaint. Version 6 addressed this by replacing territory-linked Gold income with a flat 3 Gold per player per turn. Territories now produce Investors during the INCOME phase instead of Gold. Thus, the scoring value of a territory is preserved in the final formula without creating a compounding mid-game Gold advantage for leading players.

6.2 Pacing and Downtime

Pacing averaged 2.1 in Playtest 1, indicating the game felt too slow. Sequential turns required each player to wait through three others before acting. Version 8 made INCOME and ACTION simultaneous. Playtest 3 confirmed the fix: pacing averaged 3.0 (neutral).

6.3 Turtling and Defensive Dominance

Playtest 2 identified turtling as the dominant perceived strategy. MCTS places Defensive at 10.8%. The problem is a perception gap. Human players over-invest in defence; near-optimal play identifies Aggressive and HubRusher as genuinely winning. The post-May-21 redesign targets this perception gap rather than weakening defence mechanically. Playtest 3 results show turtling was less dominant in v9, though the King's Castle's Investor income was flagged as a separate form of dominant play.

Prior changes (removing the Gold-per-Investor attack cost, adding Strike Action and Political Coup) were insufficient. The post-May-21 redesign raises the opportunity cost of holding Investors defensively: base contests increased 1→2, Angel Investors halves attacker losses, Onboarding feeds directly into contesting, Rapid Expansion doubled.

Internal playtesting found a secondary problem: Onboarding's rework removed its land-placement utility, reducing hand Investors. A proposed income increase (Hamlet 1→2, Thriving City 2→3, King's Castle 3→4) was tested and reverted. It resolved the shortage but amplified turtling: more incoming Investors made defensive stacking feel more rewarding.

6.4 First-Player Advantage

The first-player advantage identified in Playtest 1 is mitigated in Version 8. Market purchases resolve simultaneously; the earliest turn-order player wins contested buys. Turn order is randomised each round, so no player consistently benefits from going first.

6.5 Anti-Win Options

Anti-win options scored 2.6 and 2.5 across Playtests 1 and 2, the weakest-rated category. The market includes disruption tools (Arcane Recession, Tax Collection, Arcane Consultancy, Mystical Arbitrage), and immediate card play now allows same-turn use after purchase. Market availability remains luck-dependent and is a candidate for future revision.

6.6 Round Length and Density

Playtest 2 players continued to 5 rounds when given extra time, reporting 3 felt too short. However, the 5-round session ran approximately 30 minutes, double the brief's 15-minute constraint.

Extending round count is not viable. The design response is increased round density: more consequential decisions per round through higher contest frequency and more impactful cards.

6.7 Theme and Card Naming

Playtest feedback identified card naming as too generic. The design response was a dual-register naming principle: each name should read as legitimate in both corporate finance and arcane practice simultaneously. Arcane Consultancy exemplifies this: in the world of Wolves of Wand Street, retaining a mage is an ordinary procurement decision.

7. Outstanding Issues and Future Work

7.1 Unresolved Design Issues

Issue	Severity	Status
Turtling dominant strategy	High	Addressed in post-May-21 redesign: 2 base contests, Angel Investors rework, Onboarding (offensive-only), Rapid Expansion ×2. Income increase tested and reverted. Requires Playtest 3 to verify.
Anti-win options limited by market availability	Medium	Partially addressed by Hostile Negotiation counter-card and immediate card play. Full resolution pending.
Arcane Recession too powerful in Round 1	Medium	One respondent in Playtest 3 flagged land income suppression in Round 1 as disproportionately impactful. Cost increase (3→4 Gold) or a 'not playable in Round 1' restriction under consideration.
King's Castle too dominant	Medium	Confirmed by Playtest 3 (2 respondents) and MCTS (HubRusher 42% win rate). Round 1 claim is effectively uncontested. Cost increase or starting Investor adjustment under consideration.
Last-mover advantage in Round 3	Medium	Going last in Round 3 ATTACK phase confers a structural information advantage. Recognised in game design literature as a class of problem without a complexity-free solution (cf. Ferguson & Melolidakis, 1995). Documented as a known design limitation. Random turn order per round is the current mitigation.

Investor shortage	Medium	Addressed through aggressive card redesign. Playtest 3 to verify whether density increase resolves the 'too short' perception.
-------------------	--------	--

7.2 Next Steps

The post-May-21 card redesign addresses the turtling issue structurally. Playtest 3 confirmed pacing (3.0) and rules clarity (4.43) improvements. Outstanding concerns: King's Castle dominance, Investor shortage, and Arcane Recession Round 1 balance.

- Turtling: Playtest 3 showed reduced turtling; King's Castle dominance is the remaining concern.
- Card naming: Complete the dual-register naming pass across all 28 cards before final submission.
- Anti-win accessibility: Anti-win scores improved to 3.14. Arcane Recession's Round 1 power may help, though at disproportionate cost.
- Last-mover advantage in Round 3: Known unsolved problem in sequential fixed-round games (Ferguson & Melolidakis, 1995). No complexity-neutral resolution identified.

8. Card Design Notes

8.1 Draw Pile Structure

Each player's 12-card personal draw pile is fixed and does not change during the game. Bought market cards go to hand, not the draw pile. Following the post-May-21 redesign, the draw pile shifts toward offensive utility: Rapid Expansion is doubled to ×2 copies and Onboarding is reworked to feed directly into contesting rather than deployment. Redeployment remains the highest-count card, as Investor mobility underpins the contest economy. The draw pile distribution reflects the MDA principle that mechanics should reliably produce the target dynamic, which is frequent contesting.

Card	Count	Function
Petty Cash	×2	Gain 2 Gold; smooths early-game resource access for market purchases.
Rapid Expansion	×2	Grants a third contest this turn; increased from ×1 to raise frequency of high-aggression turns.
Headhunt	×1	Gain 1 Investor into hand; burst Investor access for contesting.
Minor Export	×2	Land modifier (+1 Gold/turn); transfers on capture; primary engine-building card.
Branch Office	×1	Land modifier (+1 Investor/turn); high-value if protected from capture.
Onboarding	×2	Your next contest this turn is fought as if you committed 1 additional Investor. Offensive-only; dual-use variant tested and set aside pending further data.

Redeployment	×3	Retrieve up to 2 deployed Investors to hand; core mobility card; highest count in draw pile.
--------------	----	--

8.2 Market Card Design Principles

Market cards use only Gold as their cost, following the removal of Influence in Version 6. Cards are organised into four suits reflecting strategic themes: Diamonds (military/combat), Hearts (manipulation), Spades (economy), and Clubs (arcane/burst effects). Card costs range from 2 to 6 Gold. The most powerful cards require holding Gold across multiple turns, creating a meaningful opportunity cost against spending on contesting.

Market cards are discarded permanently after use. This prevents the draw pile from accumulating powerful cards and keeps the power ceiling consistent across all three rounds.

8.3 Finance-Magic Dual Register Naming Philosophy

The card naming convention follows a dual-register principle: each name should read as a legitimate term in both corporate finance and arcane practice simultaneously, not a finance term with a magic suffix. The existing card names were evaluated against this principle and retained where they already satisfy it. Hostile Takeover, Corporate Espionage, and Political Coup all function in both registers without modification. Two cards received new names as part of the card effect redesign: Arcane Recession (replacing the retired Market Crash J of Spades) and Arcane Consultancy (replacing the retired J of Clubs AoE Gold drain), both of which commit fully to the Finance-Magic dual-register fiction.

8.4 Post-Internal-Playtest Card Changes

The following card effects were revised following internal playtesting of the post-May-21 card set. Both changes address issues observed in the internal session.

Card	Slot	Old Effect	New Effect	Cost	Rationale
Economic Dominance	A of Spades	Double Gold income for next 2 INCOME phases.	In addition to committed Investors, spend Gold to act as additional Investors for your next contest at 1 Gold = 1 Investor.	3 Gold (was 5)	Reduced cost reflects dual spending. Directly addresses Investor shortage by converting accumulated Gold into combat power.
Arcane Recession	J of Spades	Target player loses half their Gold (rounded down). [RETIRED]	Target land does not provide INCOME (including modifiers) until the land is captured by another player.	3 Gold	Retired the Gold-halving effect. New effect attacks the income engine directly

					and incentivises aggression toward suppressed lands.
Arcane Consultancy	J of Clubs	All other players lose 2 Gold immediately. [RETIRED]	Remove 2 Investors from a target land back to the Investor pile.	3 Gold	Retired the weak AoE Gold drain. Arcane Consultancy is the fireball card: targeted Investor removal that commits fully to the Finance-Magic dual-register fiction.

9. Reflection on Group Processes

9.1 Summary of Production Processes

The team adopted an iterative design process. Each version was preceded by at least 2 rounds of internal playtesting (some reaching 3–4) before external exposure, allowing obvious mechanical problems to be resolved before collecting external feedback.

Version control was maintained through shared ruleset documents, with each major iteration saved distinctly. The playingcards.io implementation was updated in parallel, providing a testable digital prototype throughout.

External playtesting used the COMP2150 Playtest Form distributed immediately after each session. A Monte Carlo Tree Search simulation was developed and run against Version 8 as computational validation. Communication and task coordination were managed through Discord with bi-weekly meetings.

9.2 Allocation of Tasks

Task	Assigned Member/s
Project Management	James (Github repo)
Initial Design / Concept	ALL
First Paper Prototype	ALL (Physical Prototype)
Second Prototype	Louis & Aidan
Final Prototype	Louis & Aidan

Resource Systems Design	ALL
Gameplay Systems Design	ALL
Card Design	Aidan
MCTS Simulation	James & Louis
Playtest 1 Materials	Louis
Playtest 1 Execution	ALL
Playtest 1 Analysis	James
Playtest 2 Materials	ALL (Physical Prototype)
Playtest 2 Execution	Louis & Aidan
Playtest 2 Analysis	ALL
Playtest 3 Materials	Louis
Playtest 3 Execution	ALL
Playtest 3 Analysis	ALL
Document Draft: Instructions	Louis
Document Draft: Design	Aidan
Document Draft: Playtesting	Marcus
Document Draft: Appendix	James
Document Proof: Instructions	Louis
Document Proof: Design	Aidan
Document Proof: Playtesting	Marcus
Document Proof: Appendix	James
Final Group Submission	Louis

9.3 Group Retrospective

Frequent communication and a high volume of internal playtesting were the strongest aspects of the group's process. The team regularly discussed design decisions and iterated across nine versions, with internal playtesting sessions preceding each external release. This meant external playtests consistently tested stable versions rather than catching basic mechanical errors. The primary weakness was delaying concept finalisation until the latter half of the project timeline. The core structure of the game (3 rounds, fixed income, simultaneous phases) was only locked in relatively late, which compressed the time available for card design and external playtesting. As a result, card balance issues (King's Castle dominance, Investor shortage, Arcane Recession Round 1 strength) were only identified in the final external playtest. In future projects, the group would prioritise locking down the core game structure earlier, starting card design before the system is fully finalised to allow more balance iterations, and scheduling external playtests earlier in the timeline to identify systemic problems while there is still time to address them.

10. Appendix

A. Playtest Survey Questions

- On average for the past 1 month, how many hours per week do you play tabletop games?
- Since which year have you played tabletop games?
- How often do you play PvP strategic/economic games?
- Were the rules easy to understand? (1: Incomprehensible – 5: Easy to Understand)
- Were the cards easy to understand? (1: Incomprehensible – 5: Easy to Understand)
- Did you feel you were good at this game? (1: Not at all – 5: Very good)
- How would you rate the pacing of the game? (1: Too Slow – 5: Too Fast)
- Does the flow of each turn feel good or clunky? (1: Clunky – 5: Perfect)
- How do you feel about your agency/freedom as a player? (1: No freedom – 5: Too many choices)
- Did you feel like you had enough options to stop a player from winning?
- Were there any dominant strategies or mechanics that felt too powerful?
- Do you have any other comments or suggestions? [open-ended]

B. Raw Playtest Data

ID	Rules	Cards	Pacing	Flow	Agency	Anti-Win	Dominant Strategy?	Key Comment
1	3	—	2	—	3	2	Unsure	A way to fight back
2	4	—	2	—	2	4	Not enough time	Income system felt unbalanced
3	5	—	3	—	4	3	Stacking money	Less income randomness
4	4	—	3	—	2	2	Farm points	Maybe more blocks
5	5	—	1	—	1	2	Income variance	Less randomness, more options
6	5	—	3	—	4	3	No	None
7	4	—	1	—	2	2	First player advantage	Too expensive; consider rummy-style
8	1	—	1	—	4	3	Rules very hard to read	Rule summary needed; test lasted 90 min
9	4	—	2	—	3	2	No dominant strategy	More options for stopping other players
10	4	—	2	—	4	4	Stacking Investors defensively	Liked the map and card effects
8	2	5	2	2	3	3	Barely one round played (online)	Hard to understand initially

9	2	5	3	4	4	3	Idk, only 1 round	Make a quick rules summary
10	4	4	5	4	3	2	Playing tall (turtling)	See notes taken
11	5	4	3	2	3	2	Turtling defence	Needs more aggression incentives
15	5	4	3	5	3	4	Land income snowball	Some free/market cards felt underwhelming
16	2	2	3	4	3	4	Investor-centric strategy	Acceptable given card limitations
17	2	2	3	4	4	4	Mechanics pretty balanced	Card combinations were fun
18	5	5	3	4	3	3	Not really	Maybe slightly more investor income
19	3	4	3	4	3	3	Arcane Recession in Round 1	Make powerful card less accessible Round 1
20	5	5	3	5	3	4	King's Castle free Round 1 claim	Difficult to get a surplus of investors
21	4	5	2	4	4	4	Being last place (turn order)	Some cards underpowered; best: income phase
22	4	4	4	3	4	2	King's Castle too strong	Not enough investors; buildings don't give enough
23	5	5	3	4	3	3	Gold hoarding / card stacking	Economy well balanced; slightly less investors/turn
24	5	4	3	4	3	3	Defensive early, offensive late	N/A

C. Version Index

Version	Description
v0	Image-based map on PCIO; hidden Gold bid contests; adjacency rules; Influence economy
v1	Territories become cards; adjacency and Gold bids retained from v0
v2	Adjacency removed; starting resources added; both sides keep half attrition
v3	Polished full ruleset; attacker wins ties; market costs normalised
v4	Deck-building hybrid experiment; territory-as-card concept explored

v5	First 3-round version; 20-card draw pile; Playtest 1 conducted here
v6–v7	Fixed income; Influence removed; 12-card draw pile; Gold-only market; immediate card play introduced
v8	Simultaneous INCOME and ACTION phases; Attack cards face-down; Playtest 2 (May 21) conducted here
v9 (CURRENT)	Aggressive card redesign; 2 base contests; reworked market cards; post-May-21 changes

D. MCTS Simulation Parameters and Raw Results

Simulation parameters: N=500 games, MCTS rollouts=30 per decision. Run against Version 8 before post-May-21 card changes.

Strategy	Win Rate	Avg Score	Games	vs Heuristic WR	Heuristic Diff
Aggressive	37.7%	5.42	231	43.3%	−5.6pp
Defensive	10.8%	4.42	251	1.2%	+9.6pp
CardEconomy	16.6%	4.46	265	7.2%	+9.4pp
HubRusher	42.0%	5.68	245	54.3%	−12.2pp
Militarist	29.6%	5.18	250	38.8%	−9.2pp
Economist	28.8%	4.97	243	9.5%	+19.3pp
Opportunist	23.5%	5.02	255	41.6%	−18.0pp
Diplomat	13.5%	4.38	260	7.3%	+6.2pp

Metric	Value	Classification
Strategy spread (MCTS)	31.3pp	IMBALANCED
Strategy spread (Heuristic)	53.1pp	IMBALANCED
Position spread	5.8pp	OK
Avg contests per round	3.02 / 3.12 / 2.78 (R1/R2/R3)	—

E. Playtest Evidence

Internal Playtest recordings:

<https://www.youtube.com/playlist?list=PL8WQf8idUSTNZhCmlSq3zmmEy3UhtGCKb>

Physical Prototype:



F. AI Acknowledgement

Claude Sonnet 4.6 (claude.ai) was used during the preparation of this report to assist with writing and running the MCTS simulations, and the drafting of this document. All decisions, analysis, card design and art, rules, and themes were written on our own. We reviewed and edited all content for accuracy and take full responsibility for this submission. This is disclosed in accordance with Macquarie University's Responsible and Ethical Use of Artificial Intelligence Policy.